Clinical Practice Guideline: Renal (RFT) & Liver Function Tests (LFT)

1. Serum Creatinine, Blood Urea, and eGFR (Kidney Function)

Serum Creatinine measures renal filtration. Normal adult creatinine is 0.7 to 1.3 mg/dL in men and 0.6 to 1.1 mg/dL in women. Normal Blood Urea is 15 to 40 mg/dL. An elevated Serum Creatinine (> 1.4 mg/dL) or reduced eGFR (< 60 mL/min/1.73m2) indicates impaired kidney filtration. In diabetic patients, persistent elevation suggests Diabetic Nephropathy.

2. Nephroprotection & Contraindications in Renal Impairment

Patients with Serum Creatinine > 1.5 mg/dL or eGFR < 60 must strictly avoid nephrotoxic painkillers such as NSAIDs (Ibuprofen, Diclofenac, Naproxen), maintain hydration, restrict excessive dietary sodium, and have antidiabetic drug dosages adjusted by a doctor.

3. Liver Enzymes: ALT (SGPT) and AST (SGOT)

ALT (normal 7 to 56 U/L) and AST (normal 8 to 33 U/L) are markers of hepatocellular integrity. Mild-to-moderate elevations (60 to 150 U/L) commonly indicate Non-Alcoholic Fatty Liver Disease (NAFLD), statin-induced transaminitis, or alcohol consumption.

Organ Marker	Normal Reference Range	Clinical Significance of Elevation
Serum Creatinine	0.7 - 1.3 mg/dL	Reduced kidney filtration / Nephropathy
eGFR	>= 90 mL/min/1.73m2	< 60 indicates Chronic Kidney Disease
Blood Urea	15 - 40 mg/dL	Dehydration or renal impairment
ALT (SGPT)	7 - 56 U/L	Hepatocellular inflammation / Fatty liver
AST (SGOT)	8 - 33 U/L	Liver or muscle tissue injury